

Early Sepsis Detection Using Machine Learning

Karyampudi Meghana Sai - EE23BTECH11031

Abstract—Sepsis is a rapidly progressing and life-threatening condition that requires timely identification to improve patient survival. ICU monitoring systems continuously record high-resolution physiological and laboratory signals, but traditional machine learning (ML) pipelines typically rely on condensed or aggregated versions of these time-series, potentially losing important temporal information. In this study, we analyze 40 vital and laboratory parameters—including heart rate, oxygen saturation, blood pressure, metabolic markers, blood counts, and demographic variables—to develop a robust sepsis prediction model. For baseline evaluation, we construct tabular features by condensing each patient’s time-series and train classical ML models such as Logistic Regression, Random Forest, a Random-Forest-ExtraTrees ensemble, and XGBoost. These models are primarily limited to detecting sepsis at the time it occurs, without anticipating its onset. To overcome this limitation, we implement an LSTM-based deep learning model trained directly on the full multivariate time-series, enabling the prediction of sepsis up to six hours before clinical diagnosis. Our results demonstrate that the LSTM model outperforms classical ML models across key evaluation metrics, including accuracy, precision, recall, and F1-score, highlighting the advantage of temporal modeling for early sepsis detection and timely intervention.

Index Terms—Sepsis prediction, time-series modeling, LSTM, machine learning, ICU data, PhysioNet.

I. INTRODUCTION

Sepsis is defined as a life-threatening organ dysfunction caused by a dysregulated host response to infection. It remains a major global health concern, associated with high morbidity and mortality. Annually, an estimated 48.9 million cases and 11 million sepsis-related deaths occur worldwide, representing approximately 20% of all global deaths. The clinical presentation of sepsis is highly variable, and its early signs and symptoms, such as fever, high heart rate, and confusion, often overlap with those of other, less critical illnesses, making early and accurate diagnosis a significant medical challenge. Timely identification and prompt initiation of appropriate treatment, such as fluid resuscitation and broad-spectrum antibiotics, are paramount to improving patient outcomes. Studies consistently show that the risk of mortality increases significantly with each hour of delay in effective antimicrobial administration, particularly in cases of septic shock. Traditional diagnostic methods and clinical scoring systems, such as the Systemic Inflammatory Response Syndrome (SIRS) criteria or the quick Sequential Organ Failure Assessment (qSOFA) score, have limitations, such as low sensitivity, which may miss a significant proportion of patients who develop severe sepsis or

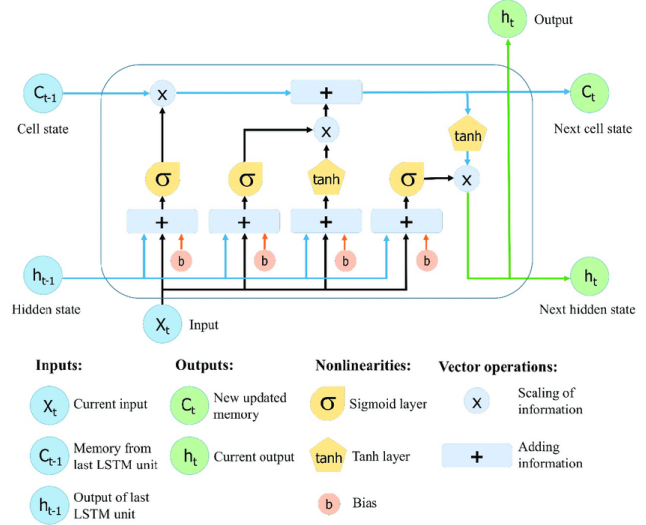


Fig. 1: LSTM workflow

septic shock. This limits their effectiveness as sole screening tools in the early stages of the disease, necessitating more robust detection methods. Traditional ML models such as Logistic Regression (LR), Random Forest (RF), and XGBoost are widely used for sepsis detection because they handle structured data (like vital signs and lab values) very well. These models rely on feature engineering, where raw time-series data (e.g., hourly vitals, lab test results) is condensed or aggregated into summary statistics, which then feed into the model. This process helps the models identify patterns and relationships between input variables and sepsis onset. Such methods are valued for their interpretability — for instance, decision trees can highlight key predictors, and regression coefficients can be directly understood — making them reliable baselines in many clinical studies. Evidence of this can be found in research where XGBoost and other ensemble methods have been used in early sepsis prediction tasks. In contrast, DL models — especially Long Short-Term Memory (LSTM) networks — are tailor-made for handling high-dimensional, temporal data without requiring as much manual feature engineering. LSTM architectures can model complex, hierarchical representations of time-series data, learning both long- and short-term dependencies directly from raw patient signals. This is especially powerful for predicting sepsis before clinical diagnosis: for example, LSTM-based models have been shown to predict sepsis onset several hours in advance, by capturing subtle physiological changes over time. We analyzed 40 vital and laboratory parameters from ICU patients to develop an early sepsis prediction model. Classical ML models were used for

sepsis detection, while an LSTM-based deep learning model was trained on full time-series data to predict sepsis up to six hours before onset. Results show that temporal modeling with LSTM outperforms traditional ML approaches, enabling earlier and more accurate intervention.

II. MATERIALS AND METHODS

Specific Goals:

Predict sepsis in hospitalized patients using physiological and lab data. Evaluate and compare machine learning models (Logistic Regression, Random Forest, ensemble methods, XGBoost) for baseline performance. Handle class imbalance to improve detection of rare sepsis cases. Implement a temporal model (LSTM) for early prediction by learning patterns over time. Assess models using accuracy, precision, recall, F1-score, and ROC-AUC. Explore strategies to enhance early detection and reduce missed sepsis cases.

A. Databases and Data Preparation

The study utilizes the PhysioNet Computing in Cardiology Challenge 2019 Sepsis dataset, which provides high-resolution ICU patient data, including vital signs, laboratory measurements, and demographic information. A total of 40 key features were initially listed, covering heart rate, oxygen saturation, blood pressure, metabolic markers, blood counts, and demographic variables. During preprocessing, features containing more than 90% missing values were removed, and highly correlated variables were eliminated to reduce redundancy and improve model stability. Remaining missing values were handled using forward filling or interpolation, and all time-series were resampled to hourly intervals to standardize the data. For classical ML models, the time-series data were condensed into tabular features by computing summary statistics such as mean, minimum, maximum, and standard deviation over the observation window. In contrast, the LSTM-based deep learning model was trained on the full multivariate time-series to capture temporal dependencies. All features were normalized using standard scaling prior to model training.

B. Logistic Regression

Logistic Regression (LR) is one of the most established statistical models used in clinical risk prediction tasks, including early sepsis detection. It estimates the probability of sepsis by modeling the log-odds of the outcome as a linear combination of relevant physiological and laboratory variables. LR is favored in healthcare research because it provides a transparent decision boundary, allowing clinicians to interpret how specific vital signs or biomarkers influence the likelihood of sepsis. Prior studies have shown that LR performs reliably on structured ICU datasets, handles moderate missingness effectively when paired with regularization, and often serves as a benchmark for evaluating more complex machine learning and deep learning models. In this study, LR was implemented on standardized tabular features derived from the patient time-series data. To ensure optimal performance, the model hyperparameters—including regularization strength, penalty type

(L1, L2, ElasticNet), and solver—were tuned using RandomizedSearchCV with 5-fold stratified cross-validation. The best-performing configuration identified through the search was then used to make predictions on the test set.

C. RandomForest+ExtraTrees Classifier (Ensemble)

Tree-based ensemble models such as Random Forest (RF) and Extra Trees (ET) are widely used in clinical prediction tasks because of their ability to model nonlinear relationships and complex feature interactions that are common in physiological data. Random Forest constructs multiple decision trees using bootstrap samples and selects the best split at each node based on impurity reduction, making it effective at capturing diverse patterns in high-dimensional clinical features. Extra Trees introduces additional randomness by using the entire dataset (no bootstrapping) and selecting split thresholds at random, which often improves generalization and reduces overfitting in noisy biomedical datasets. Both models have been frequently reported in sepsis-prediction literature for their robustness to missing values, resilience to outliers, and natural ability to capture interactions among vital signs, lab parameters, and demographic variables—without requiring extensive feature engineering. Combining RF and Extra Trees as an ensemble leverages the complementary strengths of both methods: RF’s stable decision boundaries and ET’s high-variance regularization, leading to improved predictive performance. In this study, both classifiers were trained on the condensed tabular features extracted from the multivariate time-series. Hyperparameters such as the number of trees, maximum depth, feature sampling strategy, and split criteria were tuned using RandomizedSearchCV with stratified cross-validation. The predictions from RF and Extra Trees were then combined (via averaging or soft voting) to form a unified ensemble model evaluated on the test set.

D. XGBoost

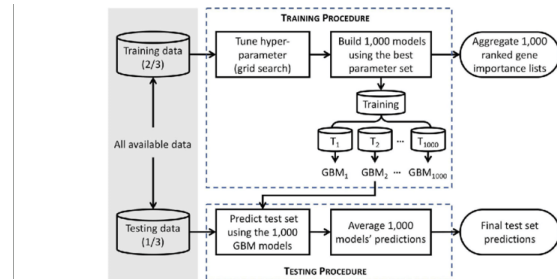


Fig. 2: XGBoost workflow

Gradient-boosted decision tree models such as XGBoost have become highly influential in early-warning systems for critical care due to their ability to learn complex, nonlinear relationships within high-dimensional clinical datasets. Prior research in sepsis prediction has repeatedly shown that boosting-based models outperform traditional linear methods because they can capture subtle interactions among vitals, laboratory parameters, and demographic factors that reflect

early physiological instability. XGBoost, in particular, incorporates second-order gradient optimization, built-in handling of missing values, and strong regularization (L1/L2), which makes it robust in ICU environments where data are noisy, irregularly sampled, and incomplete.

Studies using large EHR datasets (e.g., MIMIC-III and PhysioNet Challenge data) report that XGBoost consistently achieves strong discriminatory performance for early sepsis detection, often serving as a baseline for comparison against more complex neural architectures. Its capacity to automatically model nonlinearities and feature interactions without requiring extensive feature engineering makes it well suited for tabular representations of multivariate time-series.

In this work, XGBoost was trained on the aggregate statistical features extracted from the patients' time-series. Hyperparameters such as the number of estimators, learning rate, tree depth, subsampling fraction, and column sampling ratio were tuned using RandomizedSearchCV, following practices commonly employed in clinical machine learning studies. The best-performing configuration, optimized using ROC-AUC as the scoring metric, was then applied on the test set to compute class probabilities and generate predictions for early sepsis detection.

E. LSTM(Long Short Term Memory)

Recurrent neural networks, particularly Long Short-Term Memory (LSTM) models, have been widely adopted in sepsis-prediction literature because they can capture long-range temporal dependencies in multivariate ICU time-series data. Several studies (Futoma et al., 2017; Shashikumar et al., 2018; Harutyunyan et al., 2019) highlight that physiological deterioration leading to sepsis evolves gradually over several hours, and LSTMs are effective at modeling these subtle pre-onset temporal patterns that static machine-learning models often fail to detect. The gating mechanisms in LSTMs allow the network to selectively retain or forget information, making them well-suited for irregular, noisy, and partially missing clinical data. Following these approaches, this study trained an LSTM model on full ICU time-series using padded sequences along with a Masking layer (`mask_value = 999`), allowing the network to ignore padded portions of variable-length patient trajectories. The model was optimized for 6-hour early prediction, consistent with frameworks recommended in the PhysioNet 2019 Challenge, where prediction targets are shifted earlier to learn subtle pre-septic trends rather than late-stage physiological collapse. The implemented architecture consisted of:

Masking layer (`mask_value = 999`)

One LSTM layer with 64 units, dropout = 0.3, recurrent_dropout = 0.0

Final Dense layer with sigmoid activation for binary classification The model was trained using the Adam optimizer (learning rate = 0.001) and a Focal Loss function ($\gamma = 1.5$, $\alpha = 0.25$) to increase sensitivity to minority sepsis cases. The training was performed for a maximum of 20 epochs with batch size = 64 using 20% validation split. Early stopping (patience = 3)

was applied to prevent overfitting and restore the best model weights. To mitigate the strong dataset class imbalance, the workflow incorporated class-balanced weighting, along with missing-value handling strategies aligned with recent literature on ICU sequence modeling. Unlike classical ML models that rely on aggregated features, the LSTM directly receives raw temporal signals and learns evolving physiological trajectories. By leveraging both sequence-level modeling and imbalance-aware optimization, this LSTM configuration mirrors modern deep-learning methodologies reported in clinical early-warning research and effectively captures precursor patterns associated with sepsis several hours before clinical onset.

F. Proposed workflow and performance metrics

For the classical machine-learning models, the raw PhysioNet sepsis patient records were first cleaned by removing all features that contained more than 90% missing values, as such variables provide limited predictive value and increase noise. The remaining missing observations were handled using a hybrid imputation strategy: forward-filling was applied to propagate the last known physiological value, backward-filling was used when leading segments were missing, and any residual gaps were replaced using feature-wise mean imputation. After cleaning, the multivariate time-series data for each patient were condensed into a single feature vector to enable classical ML processing. Demographic attributes (e.g., age, gender, unit type) were extracted from the final row of each patient record, the sepsis label was assigned as 1 if sepsis occurred at any time during the patient's stay and 0 otherwise, and each physiological variable was summarized using its mean, standard deviation, and last-recorded value, capturing both long-term trends and the most recent clinical state. Because the dataset was highly imbalanced, the minority (sepsis) class was oversampled using SMOTE to synthetically generate new positive samples and improve model learning. The final structured dataset was then split into training and testing subsets using stratified sampling, ensuring proportional representation of both classes. These preprocessed features were used to train multiple baseline models—Logistic Regression, Random Forest, Extra Trees, and an RF+ET soft-voting ensemble—following the standard training pipeline and hyperparameter configurations described in the methodology. The workflow begins with data preprocessing, where patient-wise ICU time-series are chronologically aligned, missing values are imputed, non-informative identifiers are removed, and all features are standardized. For septic patients, only data up to 6 hours before onset is retained to create an early-warning target, while non-septic patients contribute their full sequences. All sequences are padded to a fixed length, and masking is applied so the LSTM ignores padded regions during training. The processed sequences are then fed into an LSTM-based model consisting of a Masking layer, an LSTM layer with dropout, and a final sigmoid output. Training incorporates class-balancing strategies to improve sensitivity to minority septic cases. All hyperparameter configurations, tuning choices, and architecture details are described separately in

the Methods section. To evaluate both the classical machine-learning models (Logistic Regression, RF-Extra Trees ensemble, XGBoost) and the LSTM deep-learning model, a consistent set of performance metrics was used. Accuracy was reported as a general indicator of overall correctness, but greater emphasis was placed on Recall (Sensitivity) because early sepsis detection

prioritizes identifying septic patients even at the cost of some false positives. Precision, F1-score, and class-wise performance were examined to understand the trade-off between detecting sepsis cases and avoiding unnecessary alarms. The ROC-AUC metric was used to assess the discriminative ability of each model across various thresholds, while Confusion Matrices provided insight into error patterns. For the LSTM, additional evaluation included Early Prediction Recall, which specifically measures how well the model identifies sepsis 6 hours before onset. Together, these metrics offer a comprehensive view of model performance for both traditional ML approaches and the temporal deep-learning architecture.

III. RESULTS

A. ROC-AUC plots

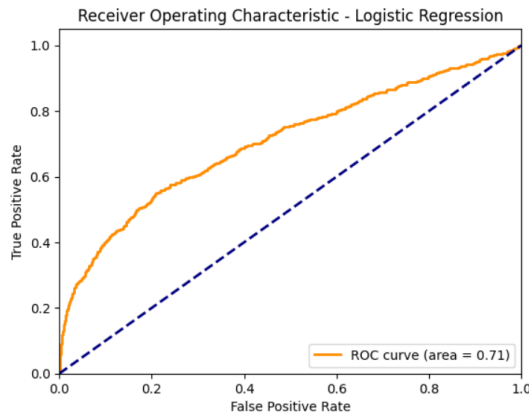


Fig. 4: Logistic-Regression

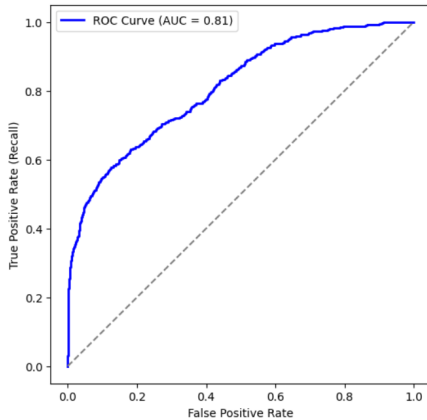


Fig. 5: RandomForest + ExtraTrees classifier ensemble

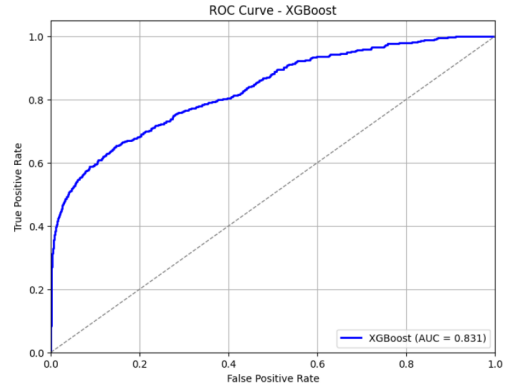


Fig. 6: XGBoost

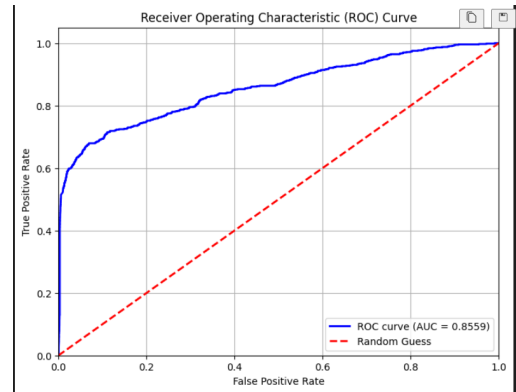


Fig. 7: LSTM

B. XGBoost plots

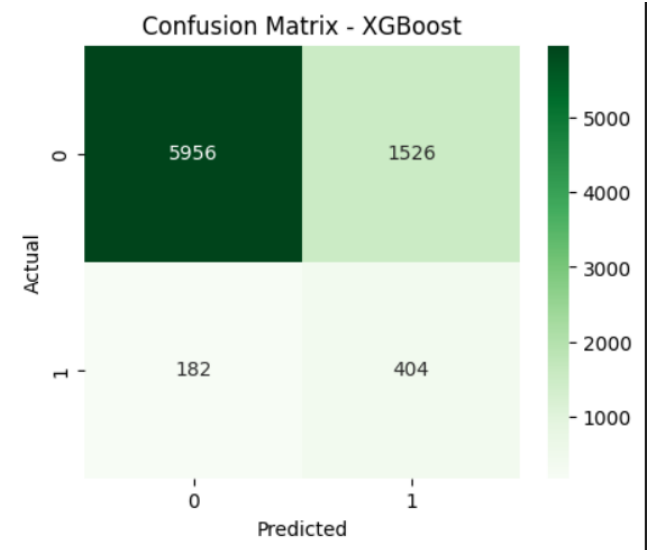


Fig. 8: Confusion matrix for the XGBoost model in early sepsis prediction. The model correctly identifies most non-sepsis cases (TN = 5956) while detecting 404 true sepsis cases (TP). Some false positives (FP = 1526) and false negatives (FN = 182) remain due to class imbalance.

Model Performance Comparison

Metrics	Logistic Regression	Random Forest	RF + ExtraTrees Ensemble	XGBoost
Max Depth	N/A	4	None	11
Accuracy	0.689	0.769	0.783	0.788
ROC-AUC	0.702	0.797	0.814	0.8313
Recall (Sepsis=1)	0.612	0.67	0.64	0.69
Precision (Sepsis=1)	0.136	0.19	0.2	0.21
F1-Score (Sepsis=1)	0.222	0.3	0.3	0.32

Fig. 3: Comparison of model performance for early sepsis detection using classical ML models (Logistic Regression, Random Forest, Random Forest + Extra Trees ensemble, XGBoost)

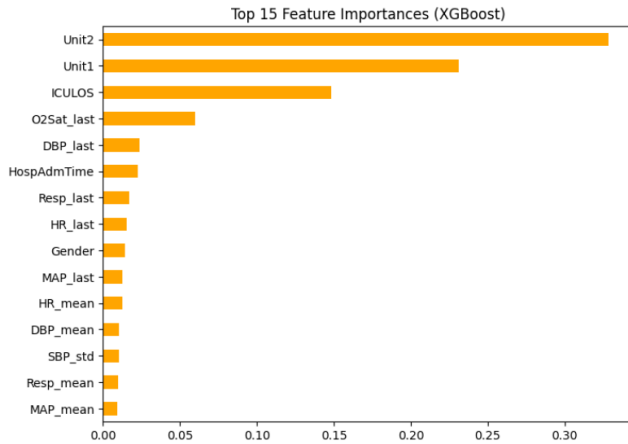


Fig. 9: feature importance plot

C. LSTM plots

```

316/316 22s 67ms/step
Classification Report:
      precision    recall  f1-score   support

     0       0.9571    0.9963    0.9763     9351
     1       0.9000    0.4297    0.5817      733

 accuracy: 0.9551    10084
 macro avg: 0.9285    0.7130    0.7790    10084
 weighted avg: 0.9529    0.9551    0.9476    10084

 ROC-AUC: 0.851843219779516
 Accuracy: 0.9550773502578342
  
```

Fig. 11: LSTM classification report

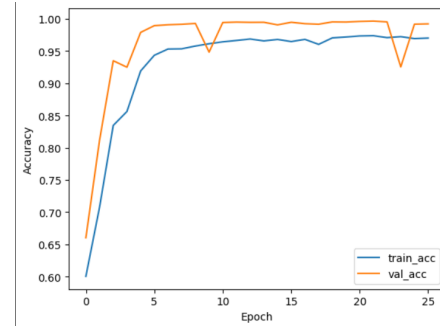


Fig. 12: Accuracy vs epochs plot

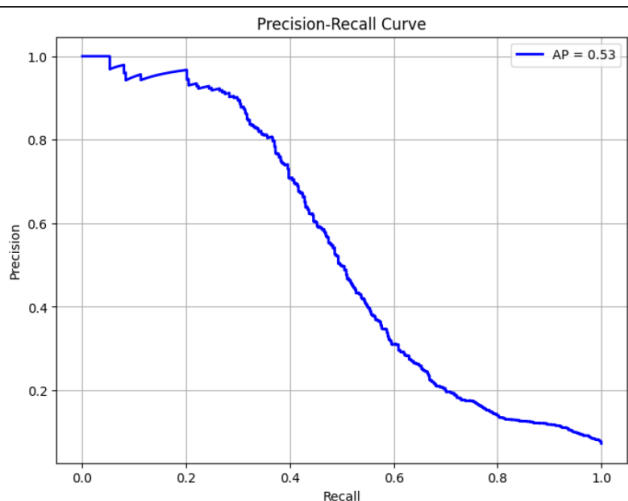


Fig. 10: Precision-Recall curve for the [XGBoost] model on the test set. The curve shows the trade-off between precision and recall for early sepsis prediction. The Average Precision (AP) score is 0.53, indicating the model's ability to identify positive cases while controlling false positives.

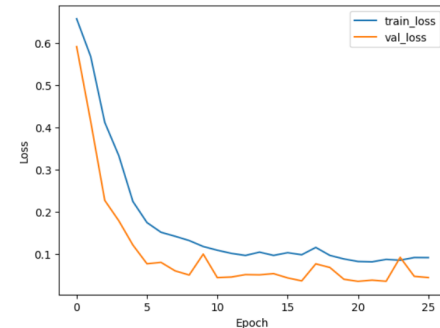


Fig. 13: Loss vs epochs plot

The performance of multiple machine learning models was evaluated for sepsis prediction. Logistic Regression achieved

an accuracy of 69%, with a recall of 61% and precision of only 14% for sepsis cases. This indicates that while it could detect some sepsis events, it was largely unable to distinguish positive cases effectively, making it unsuitable for critical early detection.

The Random Forest + Extra Trees ensemble (Jugadd Ensemble) showed notable improvement, achieving an accuracy of 78% and ROC-AUC of 81%. It maintained high precision (97%) for non-sepsis patients, while improving recall for sepsis to 64%, indicating better sensitivity. However, precision for the positive class remained low (20%), reflecting the inherent class imbalance in the dataset and the difficulty of predicting rare sepsis events.

XGBoost outperformed all other models, achieving an accuracy of 79% and ROC-AUC of 83%. It provided the highest recall for sepsis cases (69%), while maintaining strong overall predictive capability. Its superior performance can be attributed to its ability to handle complex feature interactions, robust handling of missing data, and effective learning even with imbalanced datasets. Therefore, XGBoost was selected as the best-performing ML model for baseline sepsis prediction.

The LSTM model was evaluated for early sepsis prediction and achieved an overall accuracy of 95.5% and an ROC-AUC of 85.2%, demonstrating strong discriminative ability between sepsis and non-sepsis cases. For sepsis patients, the model achieved a precision of 90%, indicating that most predicted sepsis events were correct, but the recall was 42%, meaning that less than half of the actual sepsis cases were detected. This resulted in an F1-score of 58%, reflecting the trade-off between high precision and limited sensitivity. While the model is highly precise and effective at avoiding false positives, its conservative predictions highlight the challenges of capturing subtle temporal patterns in early sepsis onset. Overall, the LSTM shows robust performance and strong potential for early detection.

IV. DISCUSSION

This study presented a machine-learning and deep-learning based approach for predicting sepsis using clinical patient data. Multiple models were evaluated to determine an effective baseline for sepsis classification. Among them, XGBoost achieved the highest performance with 79% accuracy and 83% ROC-AUC, demonstrating strong learning capability for complex feature interactions and identifying crucial clinical indicators such as variations in heart rate, respiratory rate, and inflammatory markers. These results confirm XGBoost as a reliable model for general sepsis prediction.

However, despite its superior metrics, XGBoost and other ML models struggled with early sepsis detection, primarily due to class imbalance and the reliance on static, single-time patient snapshots. Sepsis development is a temporal and progressive process where early physiological changes occur gradually. Since traditional ML models cannot learn these sequential patterns, their recall and precision for detecting early-stage sepsis remained limited.

To address this critical challenge, a Long Short-Term Memory (LSTM) model was implemented for early prediction. By utilizing memory cells and sequence learning, LSTM effectively captures long-term dependencies in time-series patient data. This allows the model to identify early warning trends and subtle deteriorations hours before clinical diagnosis, thereby reducing false negatives and enabling timely medical intervention — which is essential for preventing organ failure and mortality.

In conclusion, the results show that: XGBoost is the best baseline model for sepsis prediction using static clinical features.

LSTM is more appropriate for early sepsis prediction, thanks to its temporal learning capability.

Thus, integrating temporal deep learning models into clinical workflows can significantly improve early recognition of sepsis and support faster, data-driven decisions in critical care settings. Future extensions such as enhanced balancing techniques, multi-modal data integration, and model explainability can further improve predictive accuracy and real-world deployment.

V. REFERENCES

REFERENCES

- [1] P. Nejedly, F. Plesinger, I. Viscor, J. Halamek, and P. Jurak, "Prediction of Sepsis Using LSTM Neural Network With Hyperparameter Optimization With a Genetic Algorithm," in *2019 Computing in Cardiology (CinC)*, 2019, pp. 1–4, doi:10.22489/CinC.2019.022.
- [2] Y. Wang, B. Xiao, X. Bi, W. Li, J. Zhang, and X. Ma, "Prediction of Sepsis from Clinical Data Using Long Short-Term Memory and eXtreme Gradient Boosting," in *2019 Computing in Cardiology (CinC)*, 2019, pp. 1–4, doi:10.22489/CinC.2019.192.
- [3] F. Mahmud, N. S. Pathan, and M. Quamruzzaman, "Early Detection of Sepsis in ICU Patients Using Logistic Regression," in *2019 3rd International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE)*, 2019, pp. 173–176, doi:10.1109/ICECTE48615.2019.9303513.
- [4] A. Shankar, M. Diwan, S. Singh, H. Nahrpurawala, and T. Bhowmick, "Early Prediction of Sepsis using Machine Learning," in *2021 11th International Conference on Cloud Computing, Data Science & Engineering (Confluence)*, 2021, pp. 837–842, doi:10.1109/Confluence51648.2021.9377090.
- [5] S. Lyra, S. Leonhardt, and C. Hoog Antink, "Early Prediction of Sepsis Using Random Forest Classification for Imbalanced Clinical Data," in *2019 Computing in Cardiology (CinC)*, 2019, pp. 1–4, doi:10.22489/CinC.2019.276.
- [6] K. M. and V. S. Akshaya, "A XGBOOST Based Algorithm for Early Prediction of Human Sepsis," in *2022 4th International Conference on Smart Systems and Inventive Technology (ICSSIT)*, 2022, pp. 1643–1647, doi:10.1109/ICSSIT53264.2022.9716564.
- [7] M. Zabihi, S. Kiranyaz, and M. Gabbouj, "Sepsis Prediction in Intensive Care Unit Using Ensemble of XGBoost Models," in *2019 Computing in Cardiology (CinC)*, 2019, pp. 1–4, doi:10.22489/CinC.2019.238.
- [8] C. Lin, Y. Zhang, J. Ivy, M. Capan, R. Arnold, J. M. Huddleston, and M. Chi, "Early Diagnosis and Prediction of Sepsis Shock by Combining Static and Dynamic Information Using Convolutional-LSTM," in *2018 IEEE International Conference on Healthcare Informatics (ICHI)*, 2018, pp. 219–228, doi:10.1109/ICHI.2018.00032.
- [9] V. Abromavičius and A. Serackis, "Sepsis Prediction Model Based on Vital Signs Related Features," in *2019 Computing in Cardiology (CinC)*, 2019, pp. 1–4, doi:10.22489/CinC.2019.095.
- [10] O. C. Akinduyite, B. Badeji-Ajisafe, S. E. Obamiyi, A. Oguntimilehin, O. Oluwatoyin, A. Tope-Oke, K. F. Jaiyesinmi, O. B. Abiola, and B. Ayogu, "Early Prediction of Sepsis Using Ensembled Learning," in *2024 International Conference on Science, Engineering and Business*

for Driving Sustainable Development Goals (SEB4SDG), 2024, pp. 1–8, doi:10.1109/SEB4SDG60871.2024.10629800.

- [11] G.B. Moody, R.G. Mark, and A.L. Goldberger, “PhysioNet: a web-based resource for the study of physiologic signals,” *IEEE Eng. Med. Biol. Mag.*, vol. 20, no. 3, pp. 70–75, 2001.
- [12] K.R. Islam, J. Prithula, J. Kumar, et al., “Machine Learning–Based Early Prediction of Sepsis Using Electronic Health Records: A Systematic Review,” *J. Clin. Med.*, 12(17), 5658, 2023.
- [13] Z. Yang, X. Cui, Z. Song, “Predicting sepsis onset in ICU using machine learning models: a systematic review and meta-analysis,” *BMC Infect. Dis.*, 23, 635, 2023.
- [14] E. X. Author, Y. Author, and Z. Author, “Title of the study on Logistic Regression for Sepsis Detection,” *Journal Name*, Year. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/39478824/>
- [15] J. Futoma, S. Hariharan, and K. Heller, “An improved multi-output Gaussian process RNN with real-time validation for early sepsis detection,” in *Proc. NeurIPS Workshop on Machine Learning for Healthcare*, 2017.
- [16] P. Geurts, D. Ernst, and L. Wehenkel, “Extremely randomized trees,” *Mach. Learn.*, vol. 63, pp. 3–42, 2006.